Lesson plan · for the teacher

Names that agree

Data engineering · Level 2 · Lesson 2

Time
40 min

Materials
Board and chalk; A list of ten made-up names, each typed two or three ways; Optional: a spreadsheet app, to make a drop-down list

Learning goals

- Find the ways one name gets typed.
- Tidy, then match to one master list.
- Stop new spellings with a drop-down list.

Activity: One spelling, one list

1. Warm-up (5 min): Write ten names on the board, several of them the same pupil typed differently. Ask pupils to count the pupils, then ask how a computer would count them.
2. Teach (8 min): Show the differences a computer sees: a space, a capital, أ and ة ,I and ى ,ه and ي. Explain tidy first, then match the master list.
3. Match (15 min): Groups find which names are the same pupil, agree one spelling for each and give each pupil a number.
4. Recount (7 min): Using the master list, groups recount a short made-up register and compare their count with the first one.
5. Wrap-up (5 min): Ask how the school could stop new spellings: one master list and drop-down lists. Then give the exit ticket.

Use made-up names, never your pupils' real ones. Include at least one Arabic letter variant and one name that isn't on the list at all, to show that it's set aside, not guessed.

Discussion

- Why does the computer see "Amina " and "Amina" as two pupils?
- What should we do with a name that isn't on the master list?
- Why look for repeats only after the names agree?

Exit ticket

- Write one name three ways a computer would count as three pupils, and the one spelling you'd keep.

Try it at school tomorrow
Write ten names on the board as different people typed them. Groups find which ones are the same pupil, and agree one spelling and a number for each. That's your class's master list.

Practice questions · for you

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1 **Which fix stops new spellings before they start?**

- ◯ Asking colleagues to type more carefully
- ◯ Cleaning the sheet every Friday
- ◯ A drop-down list made from the class list

2 **The class list says “Amina”. Which of these will a computer count as a different name? Choose all that apply.**

- ☐ “Amina ” (with a space at the end)
- ☐ “Amina”
- ☐ “Aminah”
- ☐ “amina”

3 **Fill the gaps so the program prints amina: no spaces at the ends, and every letter small.**

```
name = "  Amina "
clean = name._____().______()
print(clean)
```

Word bank: `upper` `split` `strip` `lower`

4 **Put the steps of cleaning a register in order.**

- ☐ Remove the true repeats
- ☐ Write down what you changed
- ☐ Trim spaces and make capitals agree
- ☐ Keep the raw sheet as it is
- ☐ Match each name to the master list

5 **This should count 2 different pupils, but it prints 3. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  names = set()
2  for name in ["Amina", "amina ", "TUNDE"]:
3      clean = name.lower()
4      names.add(clean)
5  print(len(names))
```

✎ **Who types names into your school’s sheets, and how many ways could your pupils’ names be spelled? What would a master list change?**

Check your answers

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1 **A drop-down list made from the class list**

With a drop-down list, people pick a name instead of typing it, so no new spelling can appear.

2 **“amina”; “Amina ” (with a space at the end); “Aminah”**

Only an exact copy matches. A different capital, a space at the end or one changed letter all make a new name to the computer, even though people read them all as Amina.

3

```
name = "  Amina "
clean = name.strip().lower()
print(clean)
```

strip() removes the spaces at both ends and lower() makes every letter small, so “ Amina ” becomes “amina”.

4 **Top to bottom: 4, 5, 2, 1, 3**

Work on a copy, so the raw data survives.

5 **Line 3: `clean = name.strip().lower()`**

lower() makes “amina ” small, but the space at the end stays, so “amina ” and “amina” are still two names.